The Two Ingredients of the Reallocation Mechanism

A notation reference for U22 (productivity) and U23 (target share)

The sectoral mechanism is built from two equations. U23 says where labor *wants* to go given productivity; U22 says how AI *changes* that productivity. Everything downstream is the composition of these two. This note fixes every symbol and the conventions that govern them.

1 Ingredient 1 — the target share (U23)

$$\theta_s^* \;=\; \frac{p_s^{\eta}}{\sum_r p_r^{\eta}} \;=\; \frac{e^{z_s}}{\sum_r e^{z_r}}, \qquad z_s \equiv \eta \ln p_s.$$

Reading. Given how productive each sector is, what fraction of the workforce does sector s ideally attract? It is a softmax over log-productivity.

Symbol	Name	What it denotes
θ_s^*	target share	*desired* employment share of sector s; the $*$ is target, not the actual share θ_s
s	sector index	the sector the share is computed for, one of $\{a, m, \tau, k\}$
p_s	productivity	value added per worker — what makes a sector attractive
η	chase intensity	how aggressively labor chases productivity (softmax inverse temperature); high $\eta \Rightarrow$ small edge pulls a big share
p_s^{η}	raw pull	sector s's unnormalized pulling power
r	dummy index	runs over *all* sectors inside the sum; a distinct letter from s only to avoid collision
$\sum_r p_r^{\eta}$	normalizer	total pulling power; dividing by it makes shares sum to 1
z_s	logit	log-attractiveness score, *defined* as $\eta \ln p_s$
e^{z_s}	exp. logit	identical to p_s^{η}

The identity that links the two forms. The two fractions are equal because

$$p_s^{\eta} \;=\; \left(e^{\ln p_s}\right)^{\eta} \;=\; e^{\eta \ln p_s} \;=\; e^{z_s}.$$

This is not cosmetic. In the p^{η} form, $\partial/\partial G$ is messy; in the e^z form with $z_s = \eta \ln p_s$, the derivative $\partial z_s/\partial G = \eta\, \partial(\ln p_s)/\partial G$ is clean, because ln turns the productivity *factor* of U22 into a *sum* you can differentiate term by term. The definition $z_s \equiv \eta \ln p_s$ is the bridge that makes U22 and U23 composable.

2 Ingredient 2 — productivity responds to AI (U22)

$$p_s \;=\; p_{s,t-1}\big[\,1 + \alpha_s^F F + \alpha_s^G G + \alpha_s^{FG} FG - \alpha_s^D \rho_s G\,\big]$$

Reading. The bracket is a gross growth factor, 1 + (net growth rate); each term is a contribution to productivity growth.

Symbol	Name	What it denotes
p_s	current productivity	sector s's productivity this period t
$p_{s,t-1}$	lagged productivity	last period's level; the base the bracket scales
F	FinTech index	economy-wide FinTech penetration
G	AI index	economy-wide AI penetration
FG	joint digital	the product $F \cdot G$, the FinTech×AI complementarity
ρ_s	routine intensity	share of sector s's tasks that are automatable — its displacement exposure
α_s^F	FinTech coeff.	strength of the FinTech boost in sector s
α_s^G	AI-boost coeff.	strength of the AI productivity boost in sector s
α_s^{FG}	interaction coeff.	extra productivity from having both together
α_s^D	displacement coeff.	strength with which automation *subtracts* productivity in s

The coefficient naming convention. On every α the **superscript names the channel** and the **subscript names the sector**:

$$\alpha \underbrace{\overbrace{{}^{F,\,G,\,FG,\,D}}^{\text{channel}}}_{\text{sector}\;s}.$$

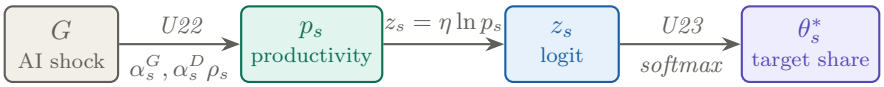
So α_k^G = "AI-boost coefficient for knowledge services," and α_m^D = "displacement coefficient for manufacturing."

Three structural facts encoded here.

1. **Only the last term is negative.** $-\alpha_s^D \rho_s G$ is the displacement drag; it scales with ρ_s, so routine-heavy sectors (manufacturing, clerical) are hit while non-routine KIS ($\rho_k \approx 0$) barely feels it.
2. **The AI boost is ordered:** $\alpha_k^G > \alpha_\tau^G > \alpha_m^G > \alpha_a^G$ — largest in knowledge services, smallest in agriculture. AI tilts the productivity map toward KIS, which is what aims the U23 target there.
3. **The net AI effect is boost minus drag:** $g_s \equiv \alpha_s^G - \alpha_s^D \rho_s$, the single object that survives differentiation, since $\partial(\ln p_s)/\partial G \approx \alpha_s^G - \alpha_s^D \rho_s = g_s$.

3 How the two compose

Differentiating θ_s^* with respect to the AI shock G runs through the chain below. Only two coefficients survive into the derivative — η from the logit definition and g_s from the U22 bracket; everything held fixed under $\partial/\partial G$ (namely F, FG, and the lag $p_{s,t-1}$) drops out.



The composite derivative is therefore

$$\frac{\partial \theta_s^*}{\partial G} = \underbrace{\frac{\partial \theta_s^*}{\partial z_\cdot}}_{\text{softmax Jacobian (U23)}} \cdot \underbrace{\frac{\partial z_\cdot}{\partial p_\cdot}}_{=\eta/p} \cdot \underbrace{\frac{\partial p_\cdot}{\partial G}}_{\text{U22 bracket}} \quad \Longrightarrow \quad \frac{\partial z_s}{\partial G} = \eta\, g_s, \quad g_s = \alpha_s^G - \alpha_s^D \rho_s,$$

and the hand-off point between the two ingredients is exactly the middle factor $z_s = \eta \ln p_s$, which converts U22's productivity response into the logit movement that U23's softmax then turns into a share.